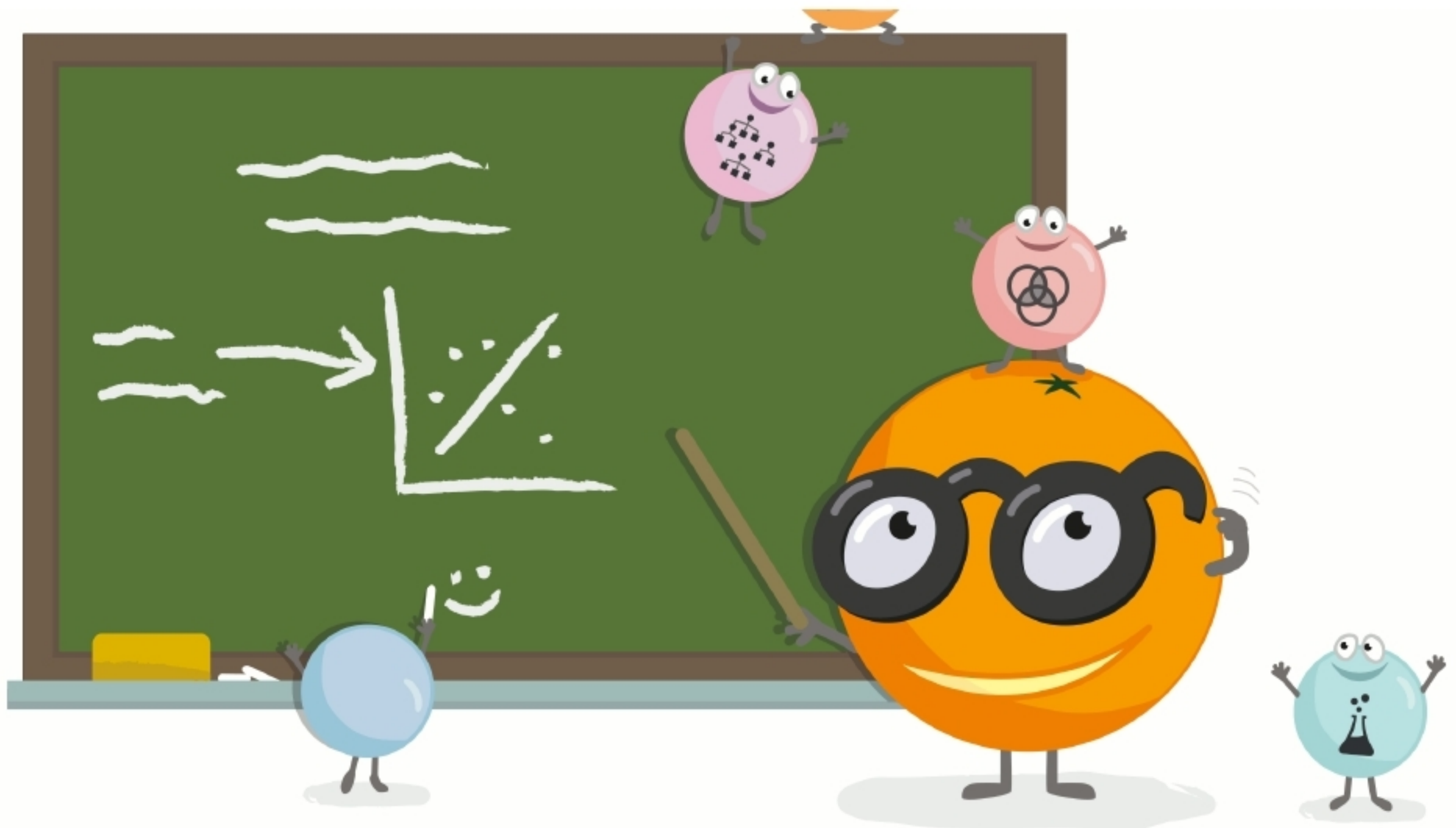


Orange: The Programming Software for Machine Learning and Data Science

The demand for machine learning engineers and data scientists has increased exponentially over the years. This is justified because machine learning is being applied in almost every field to solve real world problems. It is expected that worldwide data creation will grow to an enormous 163 zettabytes (ZB) by 2025. This article introduces Orange, a visual programming software package released under the GPL, and focused on components for data visualisation, machine learning, data mining, and analysis of data.



There are very good programming languages like Python and R to acquire the skills required for machine learning engineers and data scientists. But people need to spend a good amount of time learning these programming languages before practising data analysis techniques. Machine learning engineers and data scientists need to apply different visualisation techniques to understand the data, and test different models to get accurate results.

In this situation, having a GUI based tool that needs minimum programming can lead to good productivity. Such tools offer easy to use visual portrayals consisting of components like symbols, menus, tabs, pointers and windows, which permit users to effortlessly analyse data. This article introduces such a free and open source tool called 'Orange'. It features a visual programming frontend for explorative rapid qualitative data analysis and interactive data visualisation.

